INCH-POUND

MIL-S-44163B
27 January 1989
SUPERSEDING
MIL-S-44163A
31 October 1985

MILITARY SPECIFICATION

SHIRT, COLD WEATHER, SYNTHETIC FIBER PILE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a synthetic fiber pile shirt used as a component of the Extended Cold Weather System.

1.2 Classification. The shirt shall be of one type in the following sizes as specified (see 6.2).

Sizes		
X-Small	Medium	X-Large
Small	Large	

2. APPLICABLE DOCUMENTS

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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SPECIFICATIONS

FEDERAL

A-A-50083	-	Bag, Plastic, Folded Garment
V-F-106	-	Fastener, Slide, Interlocking
V-T-285	-	Thread, Polyester
DDD-L-20	-	Label: For Clothing, Equipage and Tentage (General Use)
PPP-B-26	-	Bag, Plastic (General Purpose)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-T-2283	-	Tape, Textile, Nylon, Woven, White or Dyed
MIL-G-16491	-	Grommet, Metallic
MIL-F-21840	-	Fastener Tapes, Hook and Loop, Synthetic
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment, Equipage; General Specification For
MIL-C-43251	-	Cloth, Pile, Synthetic Fiber Pile
MIL-C-43701	-	Cord, Elastic, Nylon
MIL-C-44162	-	Cloth, Plain Weave, Polyester/Cotton

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members For Containers and Pallets
MIL-STD-1491	-	Glossary of Knitting Imperfections

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

COLOR ASSOCIATION OF THE UNITED STATES, INC.

Department of Defense Standard Shades for Sewing Threads

(Application for copies should be addressed to the Color Associatin of the United States, Inc. East 38th Street, New York, NY 10016.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from the specification may appear in the sample, in which case the specification shall govern.

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Basic material. The basic cloth for the shirt shall be Brown 463, class 4 conforming to MIL-C-43251.

3.3.2 Pocket and reinforcement material. The cloth for reinforcing shoulders, underarms, and sleeves, and for pockets shall be Black 464, polyester/cotton, plain weave, conforming to MIL-C-44162.

3.3.3 Thread, polyester. The thread for seaming and stitching shall conform to type I, class 1, subclass A, size B of V-T-285. The thread shall be Brown C.A. 66069 and shall show good fastness to laundering.

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3.3.4 Tape, hanger. The tape for hanger loop at neck shall be flat nylon, black, 1/2 inch, conforming to MIL-T-2283.

3.3.5 Tape, fastener. The 1/2 inch fastener tapes hook and pile for pocket closure and wrist adjustment shall conform to type II, class 1, color black of MIL-F-21840.

3.3.6 Elastic cord. The drawcord shall be black, elastic nylon, type II, conforming to MIL-C-43701.

3.3.7 Barrel lock. The barrel locks for the ends of drawcords shall be black, ITW Nexus Barreloc, or equivalent (see 6.6).

3.3.8 Grommet. The grommet set used for drawcord openings shall conform to type I, class 3, size 0 of MIL-G-16491.

3.3.9 Slide fastener. The slide fastener shall be type IV, style 8, size M, black polyester or nylon continuous element coil or ladder configuration with long tab pull and thong conforming to V-F-106. The lengths in inches are as specified below:

X-Small	26-7/8
Small	27-3/8
Medium	27-7/8
Large	28-3/8
X-Large	28-7/8

3.3.10 Labels. Each shirt shall have an identification label, a size label, and an instruction label conforming to type VI of DDD-L-20, except that all labels shall have a 1/2 inch border on all sides. The label color shall be Khaki, C.A. 70188. The labels shall show fastness to laundering.

3.3.10.1 Identification label. The identification label shall conform to class 1. The item description shall be: SHIRT, COLD WEATHER, SYNTHETIC FIBER PILE.

3.3.10.2 Size label. The size label shall conform to class 2 and contain the following information for each respective size.

X-Small

Chest: Under 33 inches
Stock No. 8415-01-228-1353
NATO Size: 7080/7484

Small

Chest: From 33 to 37 inches
Stock No. 8415-01-228-1354
NATO Size: 7080/8494

Medium

Chest: From 37 to 41 inches
Stock No.8415-01-228-1355
NATO Size: 7080/9404

Large

Chest: From 41 to 45 inches
Stock No. 8415-01-228-1356
NATO Size: 7080/0414

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X-Large

Chest: From 45 inches and upward
Stock No. 8415-01-228-1357
NATO Size: 7080/1424

3.3.10.3 Instruction label. The instruction label shall conform to class 3 and contain the following information:

SHIRT, COLD WEATHER, SYNTHETIC FIBER PILE

1. Fixed Laundry: The shirts should be laundered utilizing "Natick Formula IV".

   Field Laundry: The shirts should be laundered utilizing "Formula II" of TM 10-280.

   Drying: The garments should be finished in a tumble-dryer loaded not more than 2/3 of rated capacity at lowest fabric cycle (do not exceed 90°F). Remove immediately from dryer. Avoid over-drying.

2. Machine/Hand Laundering: Use delicate/gentle fabric wash cycle or wash by hand using cold water (up to 85°F) and cold water laundry detergent. Rinse in clean cold water. NOTE: DO NOT USE CHLORINE BLEACH OR STARCH.

   DRYING: Tumble dry at lowest fabric cycle, delicate/gentle (do not exceed 90°F). Remove immediately at end of drying. Avoid over-drying. To drip dry, remove from water and place on rust-proof hanger.

3. DO NOT USE CHLORINE BLEACH, STARCH OR DRY CLEAN.

4. DO NOT PRESS.

DO NOT REMOVE THIS LABEL

3.3.10.4 Combination size, identification and instruction label. The identification label, size label and instruction label may be combined into one label, conforming to class 14. The three labels shall be printed as one continuous label with the size label first and the identification and instruction labels placed below the size label. The size and identification labels may be combined and the contents placed above the instruction label. A space of 1/2 inch minimum shall be maintained between the labels and around the borders. In addition, a solid line 1/16 inch minimum width shall extend across the entire label approximately in the middle of the 1/2 inch blank space. The coating requirement shall conform to the class 3 label.

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3.4 Design. The shirt shall be a long sleeve, front opening type with a fold over collar that converts to a turtleneck when the slide fastener is completely closed and with adjustable wrist openings, two front pockets, two hand warmer pockets, and reinforcements at the shoulders and elbows.

3.5 Patterns. Standard patterns which provide an allowance of 3/8 inch for side seams, sleeve seams, underarm inserts and collar setting, 3/8 inch allowance for sleeve tab and breast pocket flaps and 1/2 inch allowance for all other seams, except where otherwise specified, will be furnished by the Government. The Government patterns shall not be altered in any way and are to be used only as a guide for cutting the contractor's working patterns. The working patterns shall be identical to the Government patterns.

3.5.1 Pattern parts. The component parts shall be cut from the material specified below and in accordance with the pattern parts indicted.

List of pattern parts

Material	Nomenclature of parts	Cut parts
Polyester fiber pile	Front	2
	Back	1
	Sleeve	2
	Collar	1
	Hand warmer pockets	2
Polyester/cotton plain weave	Breast pockets	2
	Under arm inserts	2
	Sleeve adjusting tab	2
	Elbow patches	2
	Sleeve yoke	2
	Pocket flap	2

3.6 Construction. The construction shall conform in all respects to the requirements specified in table I and herein. Figure 1 is furnished solely for guidance and information. Should variation from this specification appear in figure 1, the document shall govern.

3.6.1 Stitches, seams, and stitching. All stitches, seams, and stitching shall conform to FED-STD-751. The type of seam, stitching, and stitches per inch shall be as specified in table I. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers, or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used. Where stitch type 401 is used, the chain or underside of the stitch shall be on the inside of the shirts.

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3.6.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/2 inch except where ends are turned under or caught in other seams or stitching. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewn.

3.6.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during end item inspection shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.6.1.2 Types 401, 502, 503, 515, 516 and 519 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.6.1.1.1a. and 3.6.1.1.1b.

3.6.2 Repairs. Repairs such as mends, darns, patches, or splices are not permitted on the shirt.

3.7 Manufacturing operations requirements. The shirt shall be manufactured in accordance with all operations specified in table I. The contractor is not required to follow the exact sequence of operations.

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TABLE I.

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cut shirts. a. Cut the shirts in strict accordance with patterns furnished which show size, directional lines, and notches for proper assembly of all parts. Parts shall be cut in accordance with marks on patterns. The pile side of the fiber pile fabric shall be on the inside of the shirt. b. Cut hem drawcords in the following lengths in inches: X-Small - 39-3/4 Large - 45-3/4 Small - 41-3/4 X-Large - 47-3/4 Medium - 43-3/4 Tolerances ± 1 c. Cut hanger tape 5-1/2 ± 1/2 inches for all sizes. NOTE: Cut draw cords and tapes with hot wire.						
2.	Replacement of damaged parts. Care shall be exercised during the spreading, cutting and manufacturing operations to assure that material defects and damages as specified in 4.4.2 are excluded and replaced with non-defective and properly matched material.						
3.	Shade and size marking. Mark, ticket or bundle all parts of the shirt to insure a uniform shade and proper assembly throughout the shirt. Any method of marking may be used except:	1					

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Shade and size marking. (cont'd)						
	a. Corrosive metal fastening devices. b. Sew-on tickets. c. Adhesive type tickets which leave an impression or traces of paper or adhesive on the material after removal of the ticket.						
4.	Make breast pockets.						
	Finished Appearance. The bellows portion of each pocket shall face toward the side seam of each shirt.						
	a. Overedge stitch sides and bottom of pocket.	502 or 503	EFd-1	8-10	B	B	
	b. Position a 3 to 3-1/4 inch strip of 5/8 inch loop fastener tape, as indicated on patterns ± 1/8 inch and stitch on all four sides 1/4 to 3/16 inch from edges.	301	LSbj-1	8-10	B	B	
	c. Fold inside and outside edges of bellows (back and bottom), as indicated on patterns and stitch 1/16 to 1/32 inch from folded edges.	301	OSf-1	8-10	B	B	
	d. Hem top of each pocket, as indicated on pattern, and stitch 1/16 to 1/8 inch from edge.	301	EFb-1	8-10	B	B	
	e. Stitch the cut-out corners of the pockets closed with two rows of stitching 1/16 to 1/32 inch apart.	301 or 401	SSa-2	8-10	B	B	
	f. Pockets may be prehemmed prior to attachment to shirt. The back and bottom edges of pocket may be turned under 1/2 inch and stitched 1/16 to 1/8 inch from edge.	301	EFa-1	8-10	B	B	

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make pocket flap.						
	a. Fold pocket flap in half with right sides together and stitch sides and corners.	301	SSae-2(a)	8-10	B	B	
	b. Turn flap right side out pushing out corners, and top stitch 3/16 to 1/4 inch from edge.	301	SSae-2(b)	8-10	B	B	
	or Heat crease 3/8 inch seam allowance under and top-stitch 1/8 inch from edge.	301	SSc-1	8-10	B	B	
	c. Position a 3 to 3-1/4 inch strip of 5/8 inch black hook fastener tape as indicated on patterns ± 1/8 inch and stitch on all four sides 1/8 to 3/16 inch from edges.	301	LSbj-1	8-10	B	B	
6.	Attach pockets and flaps to fronts.						
	a. Turn in edges of breast pocket 1/2 inch, if not prehemmed (see operation 4.f.), position pockets as indicated on patterns and stitch 1/16 to 1/8 inch from folded edge around sides and bottom, tacking tops.	301	LSd-1	8-10	B	B	
	b. Position breast pocket flaps above pockets so that the line of stitching will align with pattern marks and lay away from pocket. Stitch to fronts 3/16 to 1/4 inch from edge.	301	LSs-2(a)	8-10	B	B	
	c. Turn flap down and raise stitch 1/4 to 5/16 inch from fold, tacking ends.	301	LSs-2(b)	8-10	B	B	
	d. Overedge stitch side opening edges of hand-warmer pockets.	502 or 503	EFc-1	8-10	B	B	

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Attach pockets and flaps to fronts. (cont'd)						
	e. Turn pocket opening side of handwarmer pockets to the inside per marks on patterns and stitch 3/8 to 1/2 inch from folded edge for hem.	301	EFa-1	8-10	B	B	
	f. Position handwarmer pockets per marks on patterns, turning seam allowance under and stitch 1/8 to 3/16 inch from folded edges, backtacking at pocket openings.	301	LSd-1	8-10	B	B	
7.	Make sleeve adjusting tab.						
	a. Fold tab in half lengthwise with right sides together and stitch length and small end closed.	301	SSa-1	8-10	B	B	
	b. Turn tab right side out, pushing out point and edges, and top stitch 1/8 to 3/16 inch from the edge.	301	SSe-2	8-10	B	B	
	or						
	Heat crease 3/8 inch seam allowance under and top stitch 1/8 inch from edge.	301	SSc-1	8-10	B	B	
	c. Position a 2 to 2-1/4 inch strip of 5/8 inch hook fastener tape as indicated on patterns $\pm$ 1/8 inch and stitch on all four sides 1/8 to 3/16 inch from edges.	301	LSbj-1	8-10	B	B	
8.	Make sleeves.						
	a. Position sleeve yokes as indicated on patterns, fold under lower edge and top stitch 1/16 to 1/8 inch from edge.	301	LSd-1	8-10	B	B	

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Make sleeves. (cont'd)						
	b. Position elbow patches as indicated on patterns. Turn under upper end and outer edge and top stitch 1/16 to 1/8 inch from edge.	301	LSd-1	8-10	B	B	
	c. Prior to hemming position a 4 to 4-1/4 inch strip of 5/8 inch loop fastener tape at finished sleeve hem, beginning 1 to 1-1/4 inches from sleeve seam toward upper sleeve and stitch on all four sides, 1/16 to 1/8 inch from edges.	301	SSv-1	8-10	B	B	
9.	Assemble shirt.						
	a. With right sides together, join fronts to sleeves.	515, 516, 519	SSa-2	8-10	B	B	
	b. With right sides together, stitch underarm insert to joined fronts and sleeves per marks on pattern.	515, 516, 519	SSa-2	8-10	B	B	
	c. With right sides together, join back to sleeves 3/16 to 1/4 inch from edge.	515, 516, 519	SSa-2	8-10	B	B	
	d. Attach collar (catching size or combination label and hanger loop at center back with the loop superimposed over label).	515, 516, 519	SSa-2	8-10	B	B	
	e. With right sides together, join side seams and sleeve seam in one continuous row of stitching taking care to maintain original shape of insert with no pleats or puckers, and catching sleeve adjusting tabs at wrists and identification and instruction labels in the left side seam.	515, 519, 519	SSa-2	8-10	B	B	

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Assemble shirt. (cont'd)						
	or						
	f. As an alternate, sleeve and front and back seams may be completed first and then the insert set in.	301 or 515, 516, 519	SSa-2	8-10	B	B	
	g. Overedge stitch around periphery of the shirt, around the underarm inserts if alternate 301 stitch type is used, and around sleeve ends.	502 or 503	EFd-1	8-10	B	B	
	h. Attach eyelets through hem. Thread drawcord through eyelets and barrow locks, knotting fused ends.						
	i. Position slide fastener to shirt fronts with right sides together and stitch in place. Turn slide fastener tape towards the inside of shirt and top stitch 3/16 to 1/4 inch from edge of fold.	301	LSq-2a and LSq-2b	8-10	B	B	
	j. Turn in hem for collar, as indicated by marks on patterns, and stitch 1/8 to 3/16 inch from edge. Hem shall finish 1/2 to 3/4 inch in width.	301	EFa-1	8-10	B	B	
	k. Turn up bottom of sleeves, as indicated by marks on patterns, and stitch 1/8 to 3/16 inch from edge. Hem shall finish 7/8 to 1-1/8 inches in width.	301	EFa-1	8-10	B	B	
	l. Turn up bottom of shirt, as indicated by marks on patterns, with drawcord in the turn-up, and stitch through shirt with the stitching continued across the ends of the hem.	301	EFa-1	8-10	B	B	

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Clean shirt.						
	a. Trim thread ends and remove loose threads.						
	b. Remove spots, stains, and markings.						
	c. Remove shade tickets.						

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3.8 Finished measurements. The measurements of the finished shirt shall conform to the requirements specified in table II. All measurements shall be taken with the slide fastener closed to top of the collar and the shirt placed smooth and flat on a table.

TABLE II. Finished measurements (in inches)

	1/2 Chest inches 1/	Sleeve length inches 2/	Back Length inches 3/
X-Small	18-5/8	32-1/4	26-3/8
Small	20-5/8	33-1/4	26-7/8
Medium	22-5/8	34-1/4	27-3/8
Large	24-5/8	35-1/4	27-7/8
X-Large	26-5/8	36-1/4	28-3/8
Tolerance			
Plus	1/2	3/4	3/4
Minus	1/2	3/4	3/4

1/ With the slide fastener closed, and shirt laying flat and as smooth as possible, continue an imaginary seam line from the sleeve and front yoke seam to a point under the arm dividing the underarm insert approximately in half.

2/ Measure from center back at collar, diagonally to sleeve yoke point, to center edge of sleeve hem.

3/ Measure down center back from collar seam to bottom edge of shirt.

3.9 Workmanship. The finished shirts shall conform to the quality of product established by this specification The occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manu-facturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the visual defects specified in table III and the dimensional requirements specified in table II.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item visual examination. The end items shall be examined for the defects listed in table III. The lot size shall be expressed in units of shirts. The sample unit shall be one shirt. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE III. End item visual defects

Examine	Defect	Classification Major	Minor
Material defects	Hole, cut, tear, smash, run, or dropped stitch 1/	101	
	Needle chew	102	
	Thin or uneven area	103	
	Mend 1/	104	
	Slub (more than twice the size of the normal diameter of the yarn), float, or snag 1/		201

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TABLE III. End item visual defects. (cont'd)

Examine	Defect	Classification Major	Minor
Color and shade	Color not as specified	105	
	Variation in shade within a part or between two or more parts 2/ 3/		202
	Streaks clearly visible 3/		203
	Barre 1/		204
Cleanness	Spot or stain up to 1/4 inch in largest dimension		205
	Spot or stain more than 1/4 inch in largest dimension	106	
	Thread ends:		
	- more than three thread ends not trimmed up to 1/4 inch on outside		206
	- five or more thread ends not trimmed to 1/2 inch on inside		207
	One or more shade tickets not removed		208
	One or more size or shade markings clearly visible on outside 3/		209
Component and assembly	Any component part or required operation omitted or added (unless otherwise classified herein)	107	
	Any operation not performed as specified (unless otherwise classified herein)		210
Cutting	Any component part not cut in accordance with directional lines as indicated on patterns or in accordance with specification requirements	108	
Seams and stitching	Any open seam (except on hems)	109	
	Open seam on hems:		
	- 1/4 to 1 inch		211
	- over 1 inch	110	
	NOTE: A seam shall be considered as being open when one or more stitches joining a seam are broken, or when two or more successive skipped stitches occur.		
	Seam or stitch type not as specified	111	
	Parts of shirt badly pleated, caught or twisted in any type of stitching	112	
	Seam twisted or puckered		212
	Thread breaks (all stitch types) backstitched less than 1/2 inch		213

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TABLE III. End item visual defects. (cont'd)

Examine	Defect	Classification Major	Minor
Seams and stitching (cont'd)	Ends of all seams and stitchings produced with 301 stitch type not caught in other seams or stitchings, or backtacked less than 1/2 inch		214
	Ends of a continuous line of stitching overlapping less than 1/2 inch		215
	Raw edges:		
	-any on outside	113	
	-on inside, 1/8 to 1/2 inch (inclusive)		216
	-on inside, over 1/2 inch	114	
	Runoff, or any line of stitching not catching all plies of material to be seamed up to:		
	- up to 1/2 inch (inclusive)		217
	- more than 1/2 inch	115	
	Stitch tension:		
	-loose tension resulting in a loose seam	116	
	-tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)	117	
	Up to two stitches per inch less than specified minimum		218
	Three or more stitches per inch less than specified minimum or more than specified maximum	118	
	Stitch tension in overedging loose tension in any area:		
	- more than 1 inch but not more than 2 inches		219
	- more than 2 inches	119	
Wrist tabs	Wrist tabs missing, misplaced or unevenly matched by more than 1/4 inch	120	

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TABLE III. End item visual defects. (cont'd)

Examine	Defect	Classification Major	Minor
Hems	Omitted	121	
	Uneven in width		220
	Irregular in width:		
	- by 3/16 inch to 1/4 inch inclusive		221
	- by more than 1/4 inch	122	
Slide fastener	Puckers or pleats around fastener	123	
	Fastener top stop more than 3/4 inch from top of collar		222
	Fastener not working properly	124	
Labels	Missing, incorrect, illegible, misplaced, or not attached as specified	125	
Pocket and pocket flaps	Flap attached crookedly, i.e., distance between sides of pocket and underside of opened flap vary by more than 1/4 inch	126	
	Flap not covering front or back edge of pocket by 3/16 inch		223
	Flap not covering front or back edge of pocket by more than 3/16 inch	127	
	Handwarmer pockets unequal in width or height by more than 3/8 inch	128	
	Front edge of breast or handwarmer pockets not parallel to front edge of shirt by 3/8 inch or more		224
	Breast pockets and/or handwarmer pockets not in horizontal alignment with each other by 3/8 inch or more	129	
Fastener tape	Any tape missing, misplaced, defective or not serving intended purpose	130	
	Any length or width not as specified		225
Barrel lock	Any lock missing, defective or not serving intended purpose		226

1/ For knitted fabric imperfection defects, see MIL-STD-1491.

2/ All parts shall be examined with wales and courses running in same direction.

3/ At normal inspection distance (approximately 3 feet).

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4.4.3 End item dimensional examination. The end items shall be examined for conformance to the measurements specified in table II. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of shirts. The sample unit shall be one shirt. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container. Open and non-continuous heat sealed seams and closure of polyethylene bag, omission of vent hole in polyethylene bag.
Content	Number of unit packs per shipping container is more or less than specified.

4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each shirt shall be folded with the arms across the back and the bottom folded up over the arms to the shoulders, then placed in a close-fitting clear polyethylene bag conforming to A-A-50083 or type I or II, style 1 of PPP-B-26. The bag shall have a 1/4 inch hole located near a bottom corner to allow air to escape.

5.1.2 Commercial. Shirts shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Six shirts of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each fiberboard shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat, alternated end for end, one in length, one in width and six in depth within a shipping container. Inside dimensions of each shipping container shall approximate 25 inches in length, 15 inches in width, and 13 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.4. Shipping shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Six shirts of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW), or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, alternated end for end, one in length, one in width and six in depth within a shipping container. Inside dimensions of each shipping container shall approximate 25 inches in length, 15 inches in width, and 13 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified, in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.4.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.4.

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5.2.3 Commercial packing. Shirts, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), shirts, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallet patterns shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding means K and L or film bonding means O or P. Pallet patterns shall be number 3 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required information legibly printed or stamped in black directly on the bags across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall indicate the following information:

Stock number
Nomenclature
Size
Quantity

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The shirts are intended for use as a component of the Extended Cold Weather Clothing System.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Size (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).

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f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids.

6.5 Source of supply. A suggested source of supply for the barrel locks is:

ITW
230 Gerry Drive
Wood Dale, IL 60191

6.6 Subject term (key word) listing.

Clothing system ECWCS

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - GL
Navy - NU

Preparing activity:
Army - GL

(Project 8415-0631)

Review activities:
Army - MD
Navy - MC
DLA - CT

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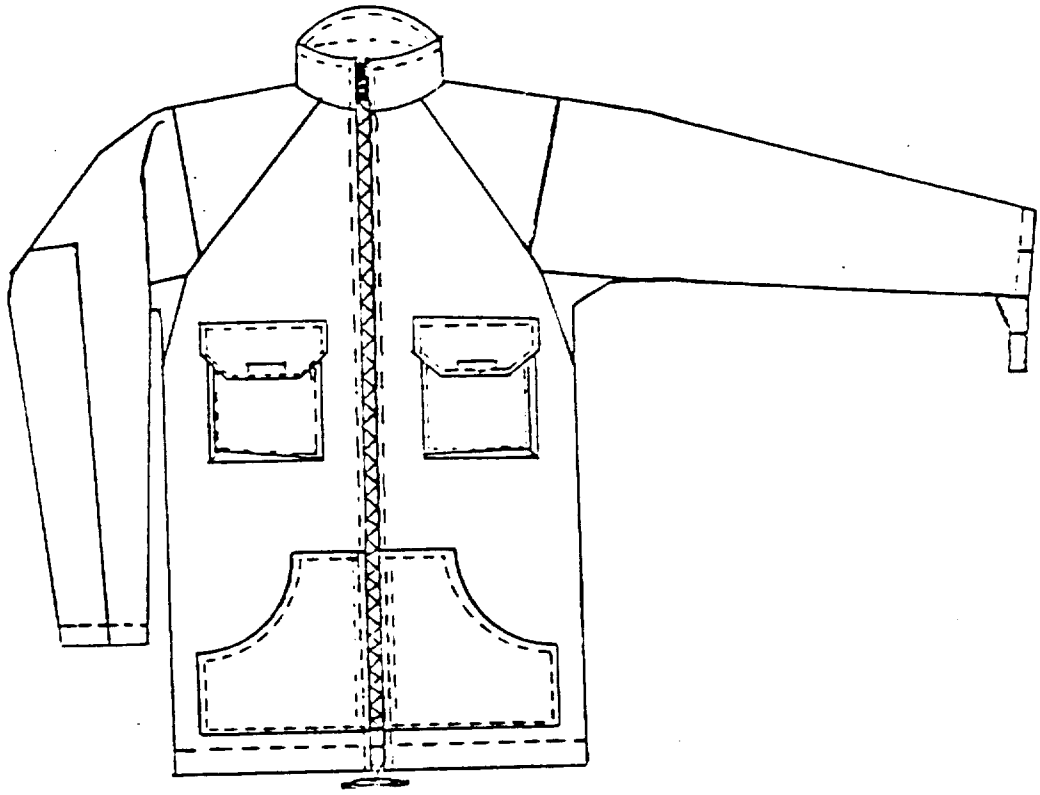


Figure 1: Shirt, Cold Weather, Synthetic Fiber Pile

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-44163B	Shirt, Cold Weather, Synthetic Fiber Pile

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 MAR

PREVIOUS EDITION IS OBSOLETE.

NATICK/OP1: Feb 86

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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